A. Serval and Mocha's Array

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, math, number theory | https://codeforces.com/problemset/problem/1789/A

Mocha likes arrays, and Serval gave her an array consisting of positive integers as a gift.

Mocha thinks that for an array of positive integers a, it is *good* iff the greatest common divisor of all the elements in a is no more than its length. And for an array of at least 2 positive integers, it is *beautiful* iff all of its prefixes whose length is no less than 2 are good.

For example:

- [3,6] is not good, because gcd(3,6)=3 is greater than its length 2.
- [1,2,4] is both good and beautiful, because all of its prefixes whose length is no less than 2, which are [1,2] and [1,2,4], are both good.
- [3,6,1] is good but not beautiful, because [3,6] is not good.

Now Mocha gives you the gift array a of n positive integers, and she wants to know whether array a could become beautiful by reordering the elements in a. It is allowed to keep the array a unchanged.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1\le t\le 500$). The description of the test cases follows.

The first line of each test case contains a single integer n ($2\le n\le 100$) — the length of array a.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($1\le a_1,a_2,\ldots,a_n\le 10^6$) — the elements of array a.

Output

For each test case, print Yes if it is possible to reorder the elements in a to make it beautiful, and print No if not.

You can output Yes and No in any case (for example, strings yEs, yes, Yes and YES will be recognized as a positive response).

Example

Input

```
6
2
3 6
3
1 2 4
3
3 6 1
3
15 35 21
4
35 10 35 14
5
1261 227821 143 4171 1941
```

Output

```
No
Yes
Yes
No
Yes
Yes
```

Note

In the first test case, neither [3,6] nor [6,3] are beautiful, so it's impossible to obtain a beautiful array by reordering the elements in *a*.

In the second test case, [1,2,4] is already beautiful. Keeping the array *a* unchanged can obtain a beautiful array.